

Ricardo Digiovanni Frumento

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PROFESSIONAL SUMMARY

PhD candidate in Computer Science at the University of South Florida specializing in robotic manipulation, multi-object grasping, and learning-based decision making. Researcher in the Robot Perception and Action Lab (RPAL) advised by Dr. Yu Sun, with experience in simulation-driven robotics, benchmarking, control, reinforcement learning, and experimental evaluation. Author of peer-reviewed publications in IEEE RA-L, IEEE T-ASE, and CoRL workshop venues, with additional experience teaching robotics, AI, and deep learning courses.

EDUCATION

University of South Florida

PhD in Computer Science

Aug 2023 – May 2027 (expected)

Relevant Coursework: Deep Reinforcement Learning, Autonomous Robots, Parallel Computing, Deep Learning

BS in Computer Science

Aug 2019 – May 2023

Relevant Coursework: Artificial Intelligence, Introduction to Robotics, Image Processing, Computational Geometry

EXPERIENCE

University of South Florida

Research Assistant, Robot Perception and Action Lab (RPAL)

Jan 2023 – Present

- Conduct research in robotic manipulation and multi-object grasping, focusing on grasp planning, scene interaction strategies, and learning-based decision making for cluttered environments.
- Design and evaluate simulation and benchmarking workflows for manipulation research, including multi-object grasping studies, scene replication, and comparative evaluation across control and learning methods.
- Develop robotics software and experimental pipelines using tools such as Python, MuJoCo, ROS, PyTorch, and OpenCV to support data collection, analysis, and system evaluation.
- Contribute to the COMPARE benchmarking ecosystem for reproducible evaluation in robotic manipulation and co-author publications in IEEE RA-L, IEEE T-ASE, and CoRL workshop venues.
- Present and discuss research findings in academic settings, collaborating with faculty and students on robotics research, experimentation, and technical documentation.

Teaching Assistant

Aug 2022 – Present

- Support upper-level computer science courses including Artificial Intelligence, Deep Learning, Robotics, and Automata Theory through office hours, grading, review sessions, and student mentoring.
- Mentor 150+ students across multiple semesters, helping reinforce technical concepts in machine learning, robotics, programming, and theoretical computer science.
- Develop assessments and instructional support materials to improve student understanding and course delivery.

Peer Leader

Sep 2021 – Aug 2022

- Facilitated collaborative study groups in introductory computer science courses, contributing to a 12% improvement in average exam performance.

SELECTED TECHNICAL PROJECTS

Gather-Grasp: Multi-Object Robotic Manipulation Pipeline

- Designed a research pipeline for studying gather-and-grasp strategies in cluttered multi-object manipulation settings.
- Evaluated action choices such as pushing, dragging, stacking, and pick-and-place to improve grasp opportunities in simulation.
- Built supporting tools for experiment configuration, data collection, and post-hoc analysis of manipulation outcomes.

COMPARE / Benchmarking for Robotic Manipulation

- Contributed to open-source benchmarking efforts aimed at improving reproducibility and standardization in robotic manipulation research.
- Helped define evaluation-oriented workflows for comparing learning and control methods across shared tasks and environments.

Simulation and Real-System Experimentation

- Developed and maintained robotics experimentation workflows bridging simulation and physical systems, including manipulation scenarios involving robotic arms, grippers, and camera-based perception.
- Worked on scene generation, object interaction analysis, and experimental tooling for grasping and manipulation studies.

PUBLICATIONS

- T. Chen, **R. Frumento**, et al. (2025). *Benchmarking Multi-Object Grasping*. IEEE Robotics and Automation Letters. doi:10.1109/LRA.2025.3595043
- Z. Ye, **R. Frumento**, Y. Sun. (2025). *Only Pick Once: Efficient Multi-Object Picking*. IEEE Transactions on Automation Science and Engineering. doi:10.1109/TASE.2025.3560956
- A. Norton, **R. D. Frumento**, et al. (2024). *Standards for Open-Source Benchmarking in COMPARE*. CoRL 2024 Workshop. OpenReview
- **R. Frumento**. (2020). *Dynamical Systems and Markov Chains*. arXiv:2012.12442

PRESENTATIONS

- **Panelist**, 2nd Bellini College REU Symposium – Spring 2026, University of South Florida, May 1, 2026
- **Presenter**, CoRL 2024 Workshop – Munich, Germany
- **Attendee**, ICRA 2025 – Atlanta, USA

TECHNICAL SKILLS

Programming: Python, C++, C, MATLAB, Fortran

Robotics / ML: PyTorch, TensorFlow, OpenCV, ROS, MuJoCo, CoppeliaSim

Tools: Git, Docker, Linux/Ubuntu

Languages: Portuguese (Native), English (Fluent), Spanish (Fluent), French (Elementary)

CERTIFICATIONS

- SQL Essential Training – LinkedIn Learning (2024)
- Artificial Intelligence Foundations: Neural Networks – LinkedIn Learning (2023)
- Responsible Conduct of Research – CITI Program (2022)